

# E-Commerce Sales Analysis and Demand Prediction Using Data Analytics and Machine Learning

Internship Project Report

## 1. Abstract

This project analyzes transaction-level e-commerce data to understand revenue patterns, product performance, geographic concentration, order-value behavior and time-based purchasing patterns. It includes data cleaning, five required visualizations, five observations, five business insights, three hypotheses, and a machine-learning component for monthly revenue prediction. The purpose is to demonstrate a reproducible data analytics workflow that can support business planning and decision-making.

## 2. Problem Statement

E-commerce businesses generate large volumes of transaction data. A structured analysis is required to identify revenue trends, high-performing products and markets, customer order behavior, and time-based patterns. The project also demonstrates whether historical transaction information can be used to estimate future monthly revenue.

## 3. Objectives

- Clean and prepare transaction data.
- Analyze revenue, products, countries and order values.
- Produce at least five meaningful visualizations.
- Document five observations and five business insights.
- Formulate three testable hypotheses.
- Build a machine-learning model for monthly revenue prediction.
- Evaluate the prediction model using MAE, RMSE and  $R^2$ .
- Provide three practical business recommendations.
- Prepare a Power BI dashboard plan and GitHub-ready project.

## 4. Dataset

The project is designed around the commonly used Online Retail transaction dataset. Typical fields include InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. Revenue is calculated as  $\text{Quantity} \times \text{UnitPrice}$ . Cancellations and non-positive sales quantities/prices are excluded from the main sales analysis.

## 5. Data Cleaning and Feature Engineering

- Convert InvoiceDate to datetime.
- Convert Quantity and UnitPrice to numeric.
- Remove duplicate transaction rows.
- Exclude cancellation invoices for the sales analysis.
- Exclude non-positive quantity and price records.
- Create  $\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$ .
- Create YearMonth, Month, DayOfWeek and Hour features.
- Use transaction-level data for descriptive analysis and monthly aggregation for revenue prediction.

## 6. Five Required Visualizations

1. Monthly Revenue Trend

2. Top 10 Products by Revenue
3. Top 10 Countries by Revenue
4. Distribution of Order Values
5. Revenue by Day of Week

The Python program generates these charts automatically in the `outputs/` folder.

## 7. Five Observations

The following are the observations to be finalized from the actual generated outputs:

1. Monthly revenue changes over time and may show seasonal or promotional patterns.
2. Revenue may be concentrated among a smaller group of products.
3. A limited number of countries may account for a large share of revenue.
4. Order values are typically unevenly distributed and may have a long tail.
5. Revenue can differ across days of the week.

Actual numerical values should be taken from the generated CSV files and charts rather than estimated.

## 8. Five Business Insights

1. Revenue trends can support inventory and campaign planning.
2. High-revenue products can receive closer merchandising and availability attention.
3. Geographic concentration can inform market-specific planning.
4. Order-value distribution can support customer segmentation and promotion design.
5. Day-of-week patterns can help schedule marketing campaigns and operational resources.

## 9. Three Hypotheses

H1: Monthly revenue changes significantly across time periods.

H2: A small number of products account for a disproportionate share of revenue.

H3: Average order value differs across countries.

## 10. Hypothesis Analysis Plan

H1 can be examined through time-series comparisons and suitable statistical comparisons of periods. H2 can be evaluated using cumulative revenue-share/Pareto analysis. H3 can be tested by comparing order-value distributions across selected countries using an appropriate statistical test. Statistical significance should only be reported after the tests are actually performed.

## 11. Machine-Learning Methodology

The machine-learning component treats monthly revenue as a regression target. Calendar/time features (year, month and a sequential time index) are used as predictors. A Random Forest Regressor is trained on an earlier chronological portion of the monthly data and evaluated on a later holdout period. This avoids randomly mixing future and past months.

## 12. Model Evaluation

The executable code generates `outputs/revenue\_prediction\_metrics.csv` containing MAE, RMSE and  $R^2$ . It also generates `revenue\_predictions.csv` and an actual-versus-predicted revenue chart. The final report should use the actual values generated by the code.

## 13. Recommendations

1. Use monthly revenue trends to plan inventory, staffing and promotional calendars.

2. Focus merchandising and availability monitoring on high-revenue products and investigate products with large revenue contribution.
3. Use country and day-of-week patterns to tailor marketing timing and geographic campaigns.

## 14. Power BI Dashboard Plan

KPI cards: Total Revenue, Total Orders, Average Order Value, Unique Products, Unique Customers.

Charts: monthly revenue trend, top products, top countries, order-value distribution and day-of-week revenue.

Slicers: country, product, month, year and customer segment where available.

Optional page: actual vs predicted monthly revenue.

## 15. Technology Stack

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Power BI and GitHub.

## 16. Limitations

The project is based on historical transaction data and therefore does not establish causation. Revenue prediction based only on calendar features is a baseline model, not a production forecasting system. Results depend on the supplied dataset and cleaning choices.

## 17. Deliverables

- Python source code
- requirements.txt
- README.md
- Project report in PDF and DOCX
- Generated visualizations and model outputs
- GitHub-ready repository structure

## 18. Conclusion

This project demonstrates an end-to-end e-commerce analytics workflow covering data preparation, visualization, observations, insights, hypotheses, business recommendations and machine learning. The final numerical results should be populated from the actual dataset execution before submission.

## 19. References

UCI Machine Learning Repository — Online Retail dataset:

<https://archive.ics.uci.edu/dataset/352/online+retail>

The Python implementation is designed to work with the commonly distributed Online Retail transaction file.